

Solution QUIZ 6 (Dated: 12-11-2024)
Data Structures and Algorithm Analysis

Roll No: _____

Name: _____

Q1. Write a recursive function to determine whether it's possible to partition an integer array into exactly two parts such that:

1. Both parts collectively include all elements of the array (with no overlap and no elements left out).
2. The sum of elements in each part is equal.

If such a partition exists, return the index of the first element in the right-hand part. Otherwise, return 0.

```
int isP(vector<int>& v, int s1, int s2, int i){
    if (s1 == s2)        return i;
    if (i == v.size())    return 0;
    return isP(v, s1 + v[i], s2 - v[i], i + 1);
}

int isP(vector<int>& v){
    int i, sum = 0;
    for (i = 1 ; i < v.size() ; i++){
        sum += v[i];
    }
    return isP(v, v[0], sum, 1);
}
```

```
input = 1, 2, 3, 4, 6
output = 0
input = 1, 1, 1, 1, 1, 2, 1
output = 4
input = 2, 3, 4, 6
output = 0
input: 1, 2, 1, 2, 2, 4, 5, 7
output= 6
```

Q2. Given an array of positive integers, write a recursive function to determine whether it is possible to divide the array into multiple sub-arrays such that each sub-array has the same sum of elements. Each sub-array must have consecutive and non-overlapping elements. You should return true if such a division is possible and false otherwise.

```
bool isMulPar(vector<int>& v, const int& pSum, int i){
    if (i == v.size())    return true;
    int j, sum = 0;
    for (j = i ; j < v.size() && sum < pSum ; j++){
        sum += v[j];
        if (sum > pSum)    return false;
        else if (sum == pSum)
            return isMulPar(v, pSum, j + 1);
    }
    return false;
}

bool isMulPar(vector<int>& v){
    int i, sum = 0;
    bool result;
    for (i = 0 ; i + 1 < v.size() ; i++){
        sum += v[i];
        result = isMulPar(v, sum, i + 1);
        if (result) return true;
    }
    return false;
}
```

```
input: 1, 4, 2, 3, 5, 5
Output: true
Input: 2, 2, 2, 2
Output: true
Input: 1, 2, 3, 4, 5
Output: false
```